

SOLUTION ARCHITECTURE REVIEW & REDESIGN

Sort money out together.

Critical evaluation · 10 launch features · Redesigned BRD
Functional requirements · Flow architecture · UI designs
Build plan you can hand to an engineering team

Version	2.0 (supersedes BRD v1.0)
Date	October 2026
Markets	Nigeria first · Ghana second
Status	Ready for build planning

START HERE

The one-page answer

MoniWise is a **strong idea with a commodity core**. Its real product is not Send and Receive (every Nigerian bank and wallet already does that well); it is the **coordination layer**: who owes what, who has paid, and why the money moved. This review keeps that insight, removes the parts of v1.0 that would slow launch or invite regulatory trouble, and turns the BRD into something an engineering team can build from.

Question	Short answer
Should this be built?	Yes , if launch is narrowed to GroupPay Pay & Collect + IOUs and non-users can pay without installing the app.
Biggest winning edge	Every GroupPay is a free invite to 3-10 people who already owe money to an existing user. Acquisition is built into the job.
Biggest flaw in v1.0	It assumes payers will use MoniWise. Most will pay by plain bank transfer, and v1.0 has no way to see or match those payments.
Single most important fix	A dedicated virtual account number per GroupPay share , so payments from any bank auto-match and auto-confirm.
What to delay	Fund Together (pooled money), Pots, merchant tools and Ghana until custody and partner permissions are signed.
Realistic launch	Closed beta month 6, public Nigeria launch month 7 with a 9-10 person core team.

How this document is organised

Part	What it covers	Read it if you are...
A. Evaluation	Scorecard, winning edge, downsides and how to fix each one	Founder, investor, PM
B. 10 launch features	What to add at launch, why it wins, how to build it, which ship Day 1	Founder, PM, design
C. Redesigned BRD	Tighter scope, business rules, KYC tiers, revenue, metrics, roadmap	Everyone
D. FRD	Module-by-module requirements with IDs and acceptance criteria	Engineering, QA
E. Flow architecture	System design, sequence flows, state machines, ledger postings, data model, APIs	CTO, engineers
F. UI designs	Nine annotated mobile screens plus the design system and UX copy rules	Design, mobile engineers
G. Build plan	Team, sprint plan, launch checklist and open decisions	Founder, PM, CTO

PART A

Critical evaluation

A1. Scorecard

Scored 1-5 against what it takes to win in Nigerian consumer payments in 2026.

Dimension	Score	Assessment
Problem clarity	●●●●●	Real, frequent and emotionally loaded ("who never pay?"). Clearly written.
Differentiation	●●●●●	GroupPay and IOU are distinctive today, but are features any big wallet can copy in a quarter.
Distribution / growth loop	●●●●●	Excellent in theory. Breaks if invitees must install an app before paying.
Regulatory realism	●●●●●	Partner-first is correct. Fund Together and Pots need sharper custody design.
Scope discipline	●●●●●	~25 P0 items for an 8-month build. Too wide for the team listed.
Business model	●●●●●	Free social payments plus per-transfer rail costs means negative unit economics until merchants or subscriptions kick in.
Technical foundations	●●●●●	Double-entry ledger, idempotency, reconciliation and provider abstraction are exactly right.
Implementation readiness	●●●●●	No data model, APIs, acceptance criteria, rounding rules, numeric SLAs or edge cases. Fixed in Parts D-E.

A2. The winning edge

Acquisition is built into the job. An organiser creating one GroupPay reaches 3-10 people who each have a reason to open the link today. That is a far cheaper acquisition channel than referral bonuses, which is how most Nigerian wallets grew.

It owns a moment, not a rail. Banks own the transfer; nobody owns "the bill just came". Owning that moment is what makes the product sticky, and it matches how people actually socialise (owambe, office lunches, trips, house bills).

Context is a data moat. Every payment carries a human reason ("David's share of dinner"). Over time this produces the richest picture of social financial behaviour in the market, which later powers risk scoring, Money Book and partner products.

Partner-first keeps capital needs low. No licence needed on day one if every regulated activity sits with a licensed partner. The team spends on product and trust, not on capital requirements.

Interoperable by default. Nigeria's instant interbank rail means a payer can use any bank app. MoniWise does not need to win the payer's wallet, only the organiser's.

Strong engineering instincts. Ledger-first, idempotent, reconciled design is what separates fintechs that survive their first fraud wave from those that do not.

A3. Downsides, risks and the fix for each

#	Problem in v1.0	Why it matters	Fix in v2.0
1	Payers are assumed to pay inside MoniWise.	Most invitees will pay with their own bank app. MoniWise never sees the payment, so status, receipts and the growth loop all break.	Virtual account per share (Feature 1). Any bank transfer auto-matches. App install becomes optional.
2	Send/Receive is presented as core.	Opening another wallet for transfers is not a reason to switch. Incumbents (OPay, PalmPay, Moniepoint, Kuda, banks) do it well and free.	Lead with GroupPay and IOU on the home screen. Send/Receive is plumbing, not the pitch.
3	Fund Together pools other people's money.	Holding third-party funds for a group is the most regulated thing in the BRD and the most fraud-prone.	Move to P1 behind a partner escrow sub-account with refund-to-source. Launch with Pay & Collect only.
4	Scope too wide for P0.	25 launch items across 8 months risks a late, shallow launch.	Re-cut P0 around the GroupPay loop (Part C3); move pooled money, Pots and merchant tools out. Later items earn their way in with data.
5	Unit economics are negative.	Each interbank transfer carries a rail cost; "free social payments" multiplies it.	Free in-app transfers, capped free external transfers, Plus subscription, VAS commissions, event and merchant fees (Part C6).
6	GroupPay is a scam vector.	Fake "contribution" groups, fake owambe links and impersonated organisers are predictable attacks.	Verified-name display, new-organiser caps, link expiry, Scam Shield (Feature 10), report button on every link.
7	Lives outside WhatsApp.	Social feed and chat are excluded (correct), but Nigerian groups coordinate on WhatsApp; v1.0 has no plan there.	WhatsApp share cards and a WhatsApp bot for reminders and status (Feature 2).
8	No rules for awkward edges.	Uneven splits (₦100,000 / 3), partial payments, overpayment, someone leaving the group, deadline passing.	Explicit business rules in Part C4: kobo rounding, overpayment handling, auto-conversion of unpaid shares to IOUs.
9	Name risk.	The "Moni-" prefix is crowded in Nigerian fintech, and app-store search will be noisy.	Run trademark and app-store clearance in month 0; hold 2 backup names.
10	Ghana treated as a second Nigeria.	Ghana is mobile-money first; the rails, KYC and partner model differ.	Ghana is a separate integration track after Nigeria proves retention.
11	Not implementable as written.	No data model, APIs, state transitions, acceptance criteria or numeric SLAs.	Parts D and E supply them.

Architect's verdict

Treat MoniWise as a **coordination product that happens to move money**, not a wallet with a split feature. Every design choice below follows from that: payers never need the app, the organiser always knows status, and unfinished obligations never disappear.

PART B

10 fintech features to add at launch

Each feature is chosen because it strengthens the GroupPay loop, builds trust, or creates a reason to return. Six can ship on Day 1. Four depend on a partner permission (e-mandate, escrow, collective savings) and are scheduled for the **launch window** (within 90 days of public launch) so they do not hold the launch hostage.

#	Feature	What it does	Why it wins	Ships
1	Pay-from-Any-Bank (virtual account per share)	Each participant gets a unique account number for their share. Transfer from any bank app; MoniWise matches it instantly.	Removes the #1 adoption barrier. Payers need no app; organisers get real-time status.	Day 1
2	WhatsApp GroupPay	Rich share card in WhatsApp; bot replies to "status", "remind" and "pay" with a link. Payment opens a light web checkout.	Meets groups where they already talk. Every message is branded distribution.	Day 1
3	Bill Snap	Photo of the restaurant bill → OCR reads total, items and service charge → assign items with a tap.	Kills the calculator. Turns item split (v1.0 Phase 2) into a 20-second task.	Day 1 (total + VAT/service), items in v1.1
4	Smart Nudges	Organiser picks a tone (Gentle, Naija-funny, Formal). MoniWise sends the reminder so the organiser is not "the one chasing". Max 1/day per person.	Removes the awkwardness, which is the emotional core of the problem.	Day 1
5	Owambe Spray	Event QR and link for weddings, birthdays and burials. Guests "spray" digital notes; celebrant sees a live wall and total.	Culturally native, high-value, extremely viral. Built on the existing Receive module.	Day 1
6	Scam Shield	Bank-verified name before every payment; warnings for first-time payee, unusual amount, new device; optional 30-min cool-off and trusted-contact alert.	Trust is the brand. Authorised-push-payment fraud is the biggest consumer fear.	Day 1
7	Bills Together	Split recurring household bills (electricity, internet, TV, rent share) and pay the biller directly once everyone has paid in.	Creates monthly habit, not just weekend use. Earns biller commission.	Launch window
8	Payday Auto-Settle	Debtor approves a direct-debit mandate: "Pay my IOU on the 28th". Automatic, polite, optional.	Raises IOU settlement rate; nobody has to ask twice.	Launch window (e-mandate partner)
9	SafeHold (social-commerce escrow)	Buyer pays into a partner-held escrow; funds release when buyer taps "Received" or after a timer.	Fixes the "what I ordered vs what I got" trust gap with Instagram/WhatsApp vendors. Paid feature.	Launch window (escrow partner)
10	Ajo Circles	Digital rotating savings: fixed members, amount, order and payout dates; contributions held by a licensed partner; auto-debit option.	Digitises a trusted habit millions already practise; strong retention.	Launch window (custody approval)

How each feature is built

#	Build approach	Partner / dependency	Effort
1	Request a dynamic account per obligation from the BaaS partner; map account → obligation; inbound credit webhook triggers ledger posting. Wrong amount: accept partial, refund excess.	Licensed bank / BaaS with dynamic virtual accounts	M

#	Build approach	Partner / dependency	Effort
2	WhatsApp Business Platform via a BSP. Template messages for invites and reminders; session replies for bot commands. Web checkout is a server-rendered page under 150 KB.	WhatsApp BSP	M
3	On-device text recognition for speed and privacy, server fallback. Parse total, VAT and service lines; user always confirms.	ML Kit / cloud OCR	S-M
4	Reminder templates + scheduler; frequency caps; stop when paid; quiet hours 21:00-08:00.	Notifications module	S
5	An "event" receive type with its own account number and QR, a live wall (WebSocket) and a settlement to the celebrant.	Same as Feature 1	S-M
6	Name enquiry on every external send; rules engine (first payee, amount vs 90-day median, new device < 24h); interstitial UI.	Rail name-enquiry via partner; risk engine	M
7	GroupPay with recurring schedule + biller payment via VAS aggregator once fully funded (needs escrow-style holding).	VAS aggregator; custody partner	M
8	Create and store mandate reference; schedule debit; handle failure with retry once then IOU stays open.	Direct-debit / e-mandate partner	M
9	Escrow sub-account per order at partner; state machine Held → Released / Disputed / Refunded.	Escrow-capable licensed partner; counsel sign-off	L
10	Circle entity + schedule; contributions into partner-held account; payout to member in rotation order.	Custody partner; counsel sign-off	L

PART C

Redesigned Business Requirements (BRD v2.0)

C1. Product definition

One-sentence definition

MoniWise helps people **sort money out together**: split a bill, collect what you are owed, keep track of who owes whom, and see where your money went. Payers can use any bank app. The organiser always knows who has paid.

Element	v1.0	v2.0 (recommended)
Hero action	Four equal actions (Send, Receive, GroupPay, Pots)	GroupPay first; Send/Receive as supporting actions; Pots after launch
Who must install the app	Implied: all participants	Only the organiser. Payers use any bank app, web checkout or WhatsApp
Launch GroupPay modes	Pay & Collect + Fund Together, equal + custom	Pay & Collect with Equal, Custom and Percentage. Fund Together in P1
Unpaid shares at deadline	GroupPay marked Expired	Unpaid shares automatically become IOUs (nothing is lost)
Positioning line	Your everyday money app	"Sort money out together." Pay. Split. Collect. Keep track.

C2. Target users and jobs-to-be-done

Persona	Job to be done	Launch feature set
The Organiser (beachhead)	"I paid; get my money back without chasing anyone."	GroupPay Pay & Collect, Smart Nudges, WhatsApp, IOU
The Payer (invitee)	"Tell me what I owe and let me pay from my own bank."	Pay-from-Any-Bank, web checkout, receipt
Flatmates / couples	"Split house bills fairly every month."	Custom/% split, Bills Together, Money Book
Celebrant / host	"Collect gifts and contributions at my event."	Owambe Spray
Young professional	"Know who owes me and where my salary went."	IOU, Money Book, Scam Shield
Micro-merchant (later)	"Get paid and confirm instantly."	Receive QR, then Merchant tools (P2)

C3. Scope: what launches and what waits

Tier	Capabilities	Gate to enter
P0 Launch	Sign-up and auth; tiered KYC; username; wallet via partner; Send (in-app + any bank) with Scam Shield; Receive (username, link, QR, account no.); GroupPay Pay & Collect (equal, custom, %); Pay-from-Any-Bank; web checkout; WhatsApp share + bot; Smart Nudges; IOU; Money Book basic; receipts and notifications; ledger + reconciliation; risk rules; admin and support console; Owambe Spray; Bill Snap (totals)	Partner contract signed; counsel opinion on wallet and collections
Launch window (+90 days)	Bills Together; Payday Auto-Settle; SafeHold escrow; Ajo Circles; Bill Snap items	Each needs its partner permission confirmed in writing
P1 Retention	Fund Together; personal Pots; saved groups; recurring GroupPay; favourites; enhanced Money Book	30-day organiser repeat rate $\geq$ 35%
P2 Business	Merchant profiles and dashboard; payment pages; invoices; API and webhooks; exports	$\geq$ 20% of volume from merchant-like accounts

Tier	Capabilities	Gate to enter
P3 Partner finance	Savings with interest, insurance, credit, cards (all partner-led)	Licensed partner + clean risk history
P4 Cross-border	Ghana (mobile-money rails), then NG-GH corridor	Nigeria unit economics positive

Still excluded from launch: loans/BNPL, crypto, investments, physical cards, payroll, POS hardware, FX accounts, international remittance, social feed.

C4. Business rules (the edges v1.0 left open)

ID	Rule
BR-01	All money is stored and calculated in kobo as integers . No floating point anywhere.
BR-02	Equal split remainder: each share = $\text{floor}(\text{total} / n)$. The remaining kobo are added to the organiser's share. Example: $\text{₦}100,000 / 3 = \text{₦}33,333.33 \times 2$ and organiser $\text{₦}33,333.34$.
BR-03	Custom and % splits must equal the total exactly before the GroupPay can be created. % shares are converted to kobo with the same remainder rule.
BR-04	Pay & Collect: participant payments go directly to the organiser's wallet . MoniWise never pools the money. This keeps launch out of collective-custody rules.
BR-05	Partial payments are accepted and reduce the share. Overpayments: the excess is refunded to source automatically within 24h.
BR-06	At the deadline (default 7 days, max 60), every unpaid or partly-paid share becomes an IOU between payer and organiser, linked to the GroupPay.
BR-07	An organiser can remove a participant only while they have paid nothing; their share is redistributed or absorbed, with a recalculation the organiser confirms.
BR-08	Reminders: max 1 per participant per day, 5 per share in total, none between 21:00 and 08:00 local time, stop automatically once paid.
BR-09	A virtual account issued for a share expires when the share is paid, cancelled or 72h after the deadline. Late transfers are refunded to source.
BR-10	New organisers (account < 30 days or Tier 1) are capped on GroupPay total and number of open GroupPays (values set by risk team).
BR-11	IOUs are records of private agreements, never MoniWise credit. MoniWise does not charge interest, penalties or late fees on IOUs.
BR-12	Refunds always go back to the original source account. Never to a different account.
BR-13	Privacy: default mode shows participants and paid/unpaid status to the group, but amounts only to the organiser and the individual payer.

C5. Identity and KYC tiers (Nigeria)

Tier structure follows the CBN three-tier KYC framework as applied by the chosen partner bank. Limits below are **placeholders** and must be replaced with the partner's current approved figures before build.

Tier	User provides	Verification	Unlocks
Guest (no account)	Nothing beyond paying from own bank	Payer's bank already did KYC; we store the sender name from the transfer	Pay a GroupPay share, spray at an event
Tier 1	Phone (OTP), name, date of birth, selfie, BVN or NIN	Identity check via KYC provider; liveness	Wallet, send/receive within low limits, create GroupPays (capped)
Tier 2	BVN and NIN both verified	Name/photo match across BVN and NIN	Higher limits, uncapped GroupPay count, Spray events
Tier 3	Address proof + ID document	Address verification; manual review if flagged	Highest limits, SafeHold seller, Ajo organiser

C6. Revenue model (unit economics fix)

Principle from v1.0 is kept: **the social loop stays free**. Revenue comes from convenience, scale and businesses. Figures are indicative starting points for pricing tests.

Stream	How it works	Indicative price	When
In-app transfers	MoniWise to MoniWise	Free (internal ledger move, no rail cost)	Launch
External transfers	To any bank account	First 10/month free, then small flat fee	Launch
Pay-in to GroupPay	Payer transfers to share account	Free to payer and organiser	Launch
MoniWise Plus	Unlimited free transfers, unlimited open GroupPays, exports, custom nudges	₦1,000-1,500 / month	Launch + 60 days
Owambe Spray	Fee on event total, paid by celebrant	1-1.5%, capped	Launch
Bills and airtime (VAS)	Commission from aggregator	Aggregator margin	Launch window
SafeHold	Fee per protected order, paid by seller	1%, capped	Launch window
Merchant QR and pages	Fee per collection	Below card rates; capped	P2
Partner referrals	Licensed savings/insurance/credit referrals	Revenue share	P3

C7. Success metrics with targets

North Star: Coordinated Money Events per Monthly Active User (kept from v1.0). Targets are for the first 90 days after public launch.

Metric	Definition	90-day target
Invite-to-pay conversion	Shares paid / shares created	≥ 70%
Time to settle	Median time from GroupPay creation to Settled	< 24 hours
Organiser repeat rate	Organisers who create another GroupPay within 30 days	≥ 35%
Payer → user conversion	Guest payers who register within 30 days	≥ 15%
Viral coefficient	New registered users generated per new organiser	≥ 0.6
IOU settlement rate	IOU value settled within 30 days of due date	≥ 60%
Payment success	Successful / attempted outbound payments	≥ 98.5%
Auto-match rate	Inbound transfers matched without manual action	≥ 99.5%
Fraud loss	Confirmed fraud loss / total volume	< 5 basis points
Support load	Tickets per 1,000 transactions	< 4

C8. Revised 12-month roadmap

Months	Focus	Exit criteria
0-1	Name clearance; company and partner shortlist; counsel opinion; UX research with 30 organisers; architecture and ledger design	Partner term sheet; validated prototype
2-3	Auth, KYC, wallet, ledger, Send/Receive, Scam Shield, admin skeleton	Internal money moves end to end in sandbox
4-5	GroupPay, Pay-from-Any-Bank, web checkout, WhatsApp, Nudges, IOU, Money Book, Spray, Bill Snap	Feature complete; security test passed
6	Closed beta: 50 friend groups, 5 events, staff and families	Invite-to-pay $\geq$ 60%, zero ledger breaks
7	Nigeria public launch	Go-live checklist (Part G3) green
8-10	Launch window features; Fund Together; Pots; Plus subscription	Organiser repeat $\geq$ 35%
11-12	Merchant beta; API foundation; Ghana partner and rails discovery	Board go/no-go on Ghana

PART D

Functional Requirements (FRD)

Each requirement has an ID, a priority and acceptance criteria written as Given / When / Then so QA can test it directly. P0 = launch, LW = launch window.

D1. Identity, access and KYC (AUTH, KYC)

ID	Requirement	Acceptance criteria	Pri
AUTH-01	Register with phone number + OTP	Given a new number, when OTP is entered within 5 min, then account is created; 5 wrong OTPs lock for 30 min.	P0
AUTH-02	Set 6-digit PIN and optional biometrics	PIN cannot be sequential or repeated digits; biometrics only unlock, PIN still required to change PIN.	P0
AUTH-03	Device binding	Login on a new device needs OTP + PIN; outbound transfers above a set amount blocked for 24h on new devices.	P0
AUTH-04	Session control	Idle timeout 5 min; user can see and revoke active devices.	P0
KYC-01	Tier 1 onboarding	BVN or NIN + selfie liveness passes → Tier 1 within 60 s in 90% of cases; failures route to manual review queue.	P0
KYC-02	Tier upgrade	User can upgrade in app; limits update immediately after approval.	P0
KYC-03	Unique username	3-20 chars, letters/digits/underscore, unique, changeable once per 90 days; old username reserved 90 days.	P0

D2. Wallet, Send and Receive (PAY)

ID	Requirement	Acceptance criteria	Pri
PAY-01	Wallet with dedicated account number	On Tier 1 approval a permanent account number is issued by the partner and shown on Home.	P0
PAY-02	Send to username / phone / bank account	External sends show bank-verified name before confirm; user must confirm with PIN or biometrics.	P0
PAY-03	Purpose and category on every send	Purpose optional, category auto-suggested; both stored on transaction and visible in Money Book.	P0
PAY-04	Idempotent payments	Same idempotency key within 24h returns the original result; double-tap never creates two debits.	P0
PAY-05	Unknown status handling	If partner times out, status = Pending, user sees "We're confirming"; request job resolves within 15 min or raises ops exception.	P0
PAY-06	Receive by link and QR	Fixed or open amount; link works in any browser; QR encodes the link.	P0
PAY-07	Payment request	Request to a user or phone; payer can pay, decline or ignore; requester sees status.	P0
SHD-01	Scam Shield checks	Given first-time payee OR amount > 3× 90-day median OR device < 24h, when user reviews, then warning screen shows with reasons and a "Cancel" primary option.	P0
SHD-02	Cool-off option	User can enable a 30-minute delay on sends above their chosen amount; cancel available until release.	P0

D3. GroupPay (GP)

ID	Requirement	Acceptance criteria	Pri
GP-01	Create GroupPay	Requires title, total, ≥ 2 participants, split method, deadline; Pay & Collect requires organiser to confirm they paid.	P0
GP-02	Split calculation	Equal / Custom / % follow BR-02 and BR-03; Create is disabled until shares sum to total; UI shows remaining difference live.	P0
GP-03	Add participants	From contacts, username, phone or "anyone with link" slots; non-users allowed.	P0
GP-04	Share account per participant	Each share gets a unique virtual account number valid per BR-09.	P0
GP-05	Auto-match inbound payment	Given a credit webhook on a share account, when amount $\leq$ outstanding, then share reduces, ledger posts, organiser is notified in < 5 s from webhook.	P0
GP-06	Overpayment	Amount above outstanding: share marked Paid, excess refunded to source within 24h, both parties notified.	P0
GP-07	Live status	Organiser and participants see status per privacy mode (BR-13); updates in real time while screen open.	P0
GP-08	Smart Nudges	Organiser selects tone; system sends via WhatsApp/SMS/push respecting BR-08.	P0
GP-09	Deadline conversion	At deadline, unpaid shares become IOUs linked to the GroupPay; status becomes "Closed - IOUs open".	P0
GP-10	Cancel	Allowed only when no payments received; otherwise organiser can close early (converts to IOUs).	P0
GP-11	Receipt	Settled GroupPay produces a shareable receipt (image + PDF) with reference, items, payers and timestamps.	P0
GP-12	Bill Snap	Photo detects total and service charge with user confirmation; confidence $< 80\%$ asks user to edit.	P0
GP-13	Fund Together	Contributions held in partner escrow; organiser releases to merchant when funded; refund-to-source if expired.	P1

D4. WhatsApp and guest web checkout (WA, WEB)

ID	Requirement	Acceptance criteria	Pri
WA-01	Share card	Share produces a WhatsApp message with title, organiser, each person's amount (per privacy) and link.	P0
WA-02	Bot commands	"status", "remind", "pay" in the GroupPay thread with the MoniWise number return the right response in < 3 s.	P0
WEB-01	Guest checkout page	Loads in < 2 s on 3G; shows organiser verified name, share, account number, timer; no login required.	P0
WEB-02	Card / USSD option	Secondary payment through the card/USSD processor; card data never touches MoniWise servers.	P0
WEB-03	Conversion prompt	After payment, show receipt and "Track your splits" install prompt; never block receipt behind install.	P0

D5. IOU, Money Book, Spray (IOU, BOOK, EVT)

ID	Requirement	Acceptance criteria	Pri
IOU-01	Record IOU	Creditor enters person, amount, reason, optional due date; debtor gets a request to confirm. Unconfirmed IOUs are labelled "Not confirmed".	P0
IOU-02	Settle IOU	Any payment linked to the IOU (in app or via its account number) reduces balance; Paid at zero.	P0

ID	Requirement	Acceptance criteria	Pri
IOU-03	Dispute IOU	Debtor can mark "I don't agree" with a note; nudges stop; both see Disputed state. MoniWise does not adjudicate private IOUs.	P0
IOU-04	Payday Auto-Settle	Debtor creates mandate with date and amount; failed debit retried once next day; never more than agreed amount.	LW
BOOK-01	Monthly view	In, Out, Net and top 5 categories for selected month; tap a category to see transactions.	P0
BOOK-02	Recategorise	User category change applies to that transaction and, if chosen, future ones from the same counterparty.	P0
EVT-01	Create Spray event	Tier 2+ user creates event with name, date, celebrant wallet; gets account number, QR and link.	P0
EVT-02	Spray	Guest taps note value, multiplier and sprays; live wall updates within 2 s; optional message (profanity-filtered).	P0

D6. Risk, operations and admin (RISK, OPS)

ID	Requirement	Acceptance criteria	Pri
RISK-01	Rules engine	Configurable rules (velocity, limits, first payee, new device, high-risk patterns) without redeploy; every decision logged with reason.	P0
RISK-02	Report link	Every GroupPay, Spray and request page has "Report"; 3 reports freeze new payments pending review.	P0
OPS-01	Reconciliation	Daily (and hourly intraday) match of ledger vs partner statements; unmatched items enter exceptions queue with SLA 4 business hours.	P0
OPS-02	Admin console	Search users, transactions, GroupPays; view audit trail; restrict account; issue refund with maker-checker approval.	P0
OPS-03	In-context support	Support opened from a transaction or GroupPay carries its IDs, amounts, timestamps and status automatically.	P0

D7. Non-functional requirements (with numbers)

Area	Requirement
Availability	99.9% monthly for payments and GroupPay APIs; 99.5% for Money Book and reporting.
Performance	API p95 < 300 ms excluding partner time; payment initiation visible to user < 2 s; webhook-to-notification < 5 s.
Scale (launch)	200 payments/second peak, 1M registered users, 10M ledger entries/month without re-architecture.
Recovery	Ledger: zero data loss (synchronous replica); RTO 1 hour. Other data: RPO 5 min.
Security	TLS 1.2+, encryption at rest, secrets in managed vault, OWASP MASVS for mobile, annual pen test, PCI scope avoided by tokenised card processor.
Privacy	Nigeria Data Protection Act alignment: consent, purpose limitation, retention schedule, data subject requests within 30 days; confirm data-residency needs with counsel.
Device reach	Android 8+ and iOS 15+; app < 40 MB; usable on 3G; crash-free sessions ≥ 99.5%.
Auditability	Every state change and admin action is append-only with actor, time, before/after.

PART E

Flow architecture

E1. System architecture

A **modular monolith**: one deployable backend with strict internal module boundaries. It is faster to build and operate than microservices, and modules can be split out later when load or team size demands it. All money movement goes through the Payments Orchestrator and is recorded by the Ledger.

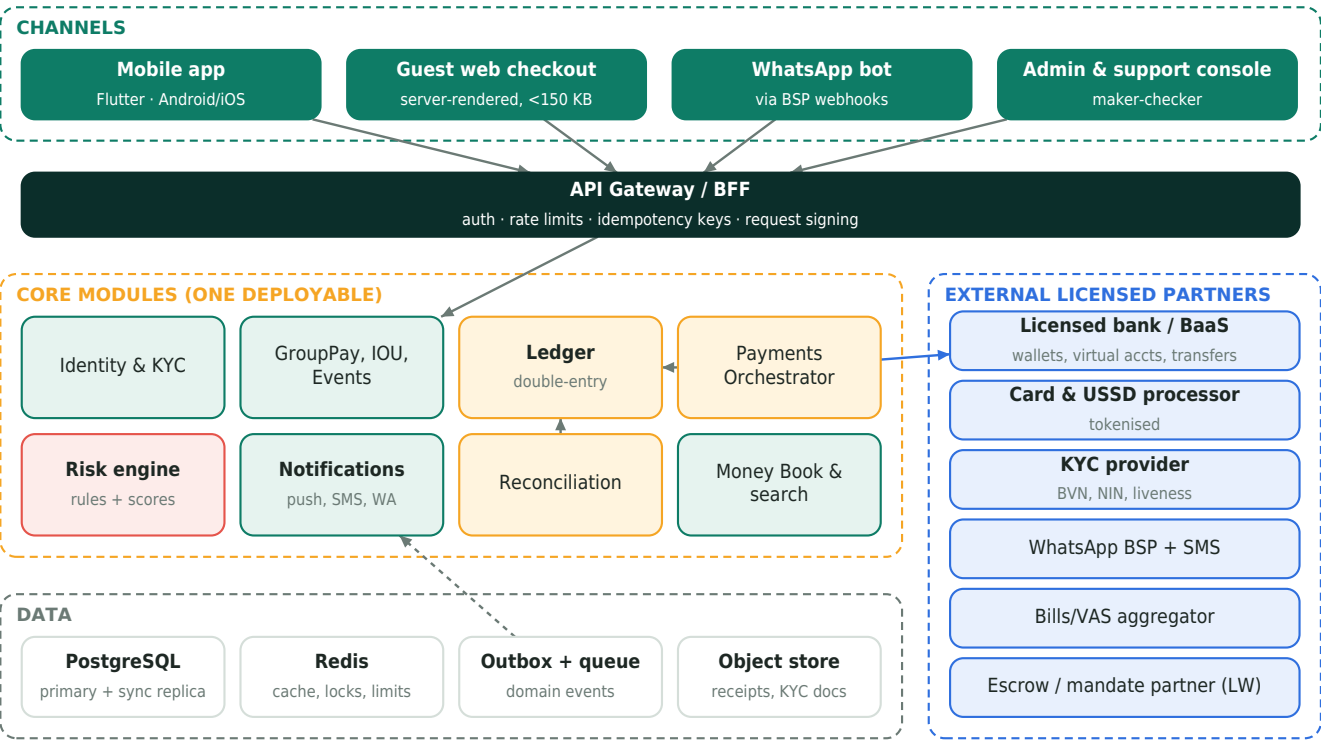


Figure 1. Logical architecture. Amber modules touch money; every amber call is idempotent and ledgered.

Layer	Recommended choice	Why
Mobile	Flutter (single codebase)	One team ships Android first (majority of NG users) and iOS together; good performance on low-end devices.
Backend	Kotlin (Spring) or TypeScript (NestJS) modular monolith	Strong typing for money code; large Nigerian talent pool for both.
Database	PostgreSQL with synchronous replica	ACID transactions for the ledger; mature tooling.
Events	Transactional outbox + managed queue	Guarantees no event is lost or sent for a rolled-back transaction.
Hosting	Major cloud region + local DR as counsel advises	Confirm any data-residency obligations for payment data before choosing.
Observability	Structured logs, metrics, tracing, alerting on ledger imbalance	Money bugs must page someone in minutes, not days.

E2. Core flow: GroupPay Pay & Collect with Pay-from-Any-Bank

This is the flow that makes or breaks the product. Note that the participant never needs the app.

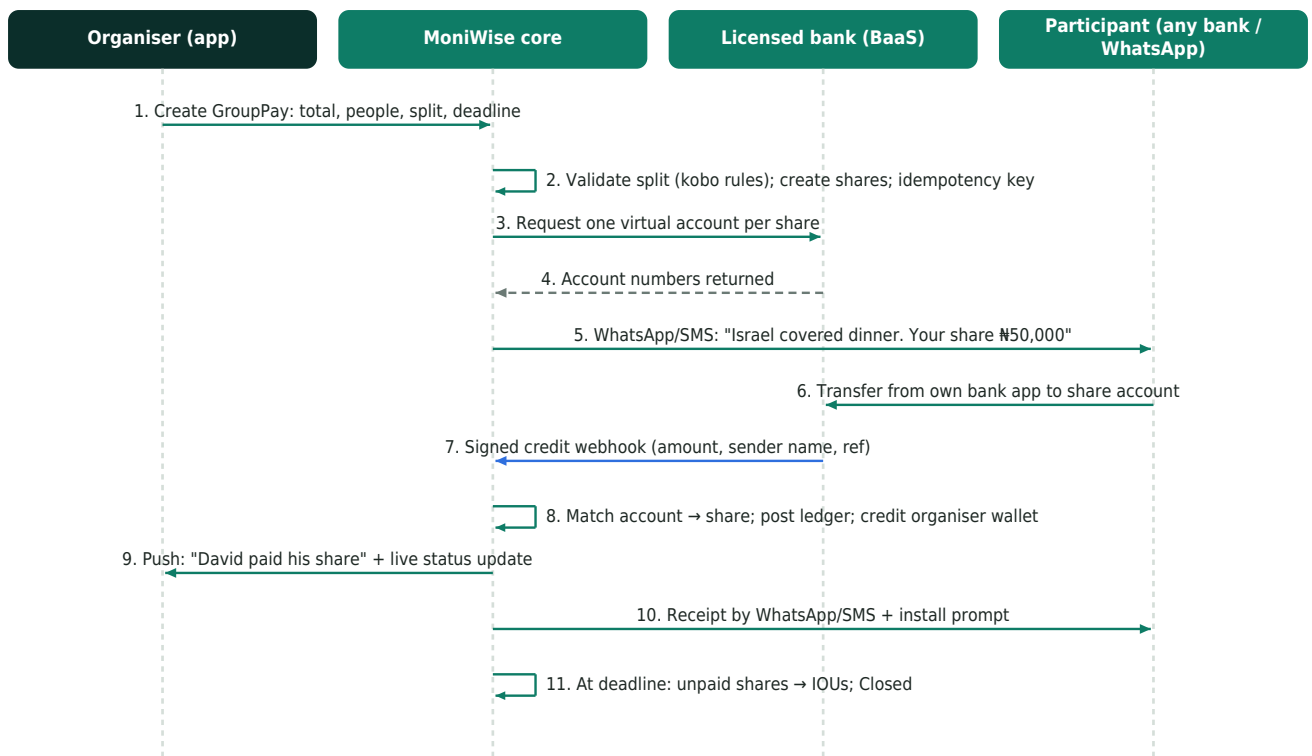


Figure 2. Webhooks are verified by signature and re-checked against the partner API before money is credited.

E3. Send flow with Scam Shield

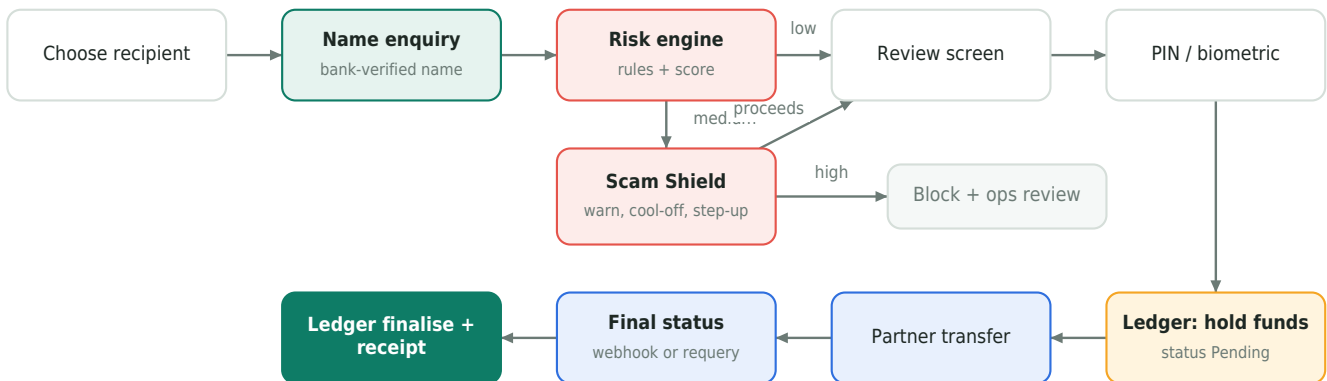


Figure 3. Failed transfers release the hold; unknown results stay Pending until requery confirms (never guessed).

E4. Domain events

Published through the outbox: **UserRegistered**, **KycTierChanged**, **PaymentInitiated**, **PaymentSucceeded**, **PaymentFailed**, **ShareCreated**, **SharePaid**, **GroupPaySettled**, **GroupPayClosed**, **IouCreated**, **IouSettled**, **SprayReceived**, **RiskDecisionMade**, **ReconciliationBreak**. Notifications, Money Book, analytics and risk subscribe to these, so new features can be added without touching payment code.

E5. GroupPay state machines

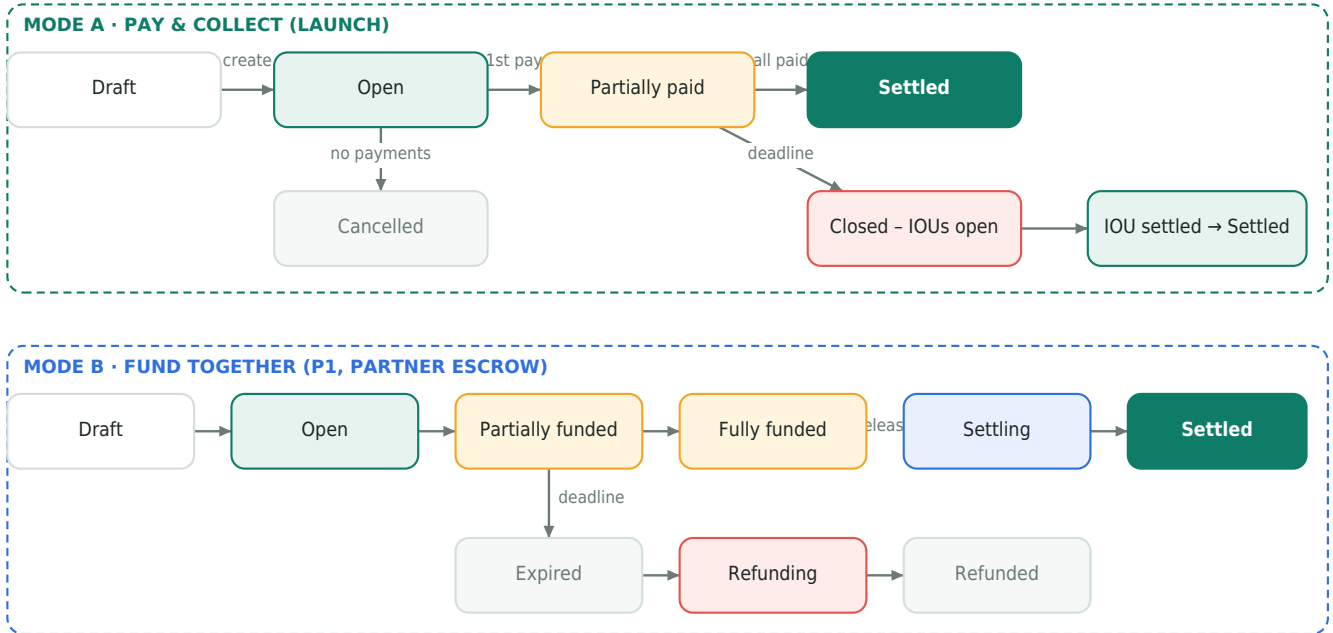


Figure 4. Only the transitions shown are legal; any other attempted transition is rejected and logged.

E6. Ledger postings: worked example

Dinner ₦250,000, five people, equal split, Israel pays the restaurant from his MoniWise wallet. Every event is a balanced journal (debits = credits).

Event	Debit	Credit	Amount
Israel pays restaurant (external transfer)	User wallet: Israel	Partner settlement (outbound clearing)	₦250,000
Transfer fee (if any beyond free quota)	User wallet: Israel	MoniWise fee revenue	₦fee
David pays share into share account	Partner settlement (inbound clearing)	User wallet: Israel	₦50,000
Tunde overpays by ₦5,000	Partner settlement (inbound clearing)	Refunds payable: Tunde's source	₦5,000
Refund sent to Tunde's bank	Refunds payable: Tunde's source	Partner settlement (outbound clearing)	₦5,000

Obligations (shares, IOUs) are **tracked separately from the ledger**. A share is a promise, not money; only real movements are journalled. Daily, the sum of all user wallet accounts must equal the balance in the partner's safeguarded pool account, otherwise an alert fires.

E7. Core data model

Entity	Key fields	Notes
User	id, phone, username, name, kyc_tier, status, created_at	Name always from verified source after KYC
Device	id, user_id, fingerprint, bound_at, trusted	Drives new-device rules
Wallet	id, user_id, partner_account_no, currency, status	Balance is derived from ledger, never stored as editable field
LedgerAccount	id, type (user_wallet, clearing, revenue, refunds_payable...), owner_id	Chart of accounts
Journal / Entry	journal_id, entry_id, account_id, debit_kobo, credit_kobo, ref_type, ref_id	Append-only; sum per journal = 0
Payment	id, idempotency_key, type, amount_kobo, status, partner_ref, purpose, category	Operational view of a money movement
GroupPay	id, organiser_id, title, total_kobo, mode, split_method, deadline, status, privacy	State machine in E5
Share	id, grouppay_id, participant (user_id or phone/name), amount_kobo, paid_kobo, virtual_account_no, status	One per participant
IOU	id, creditor_id, debtor_ref, amount_kobo, paid_kobo, due_date, status, source_grouppay_id	Confirmed / Not confirmed / Disputed
Event (Spray)	id, host_id, celebrant_wallet_id, account_no, total_kobo, starts_at, ends_at	Live wall via WebSocket
Notification	id, user_or_phone, channel, template, payload, sent_at, status	Respects quiet hours
RiskDecision	id, subject_type, subject_id, rules_fired, score, action, reviewer	Full audit trail
AuditLog	id, actor, action, target, before, after, at	Append-only

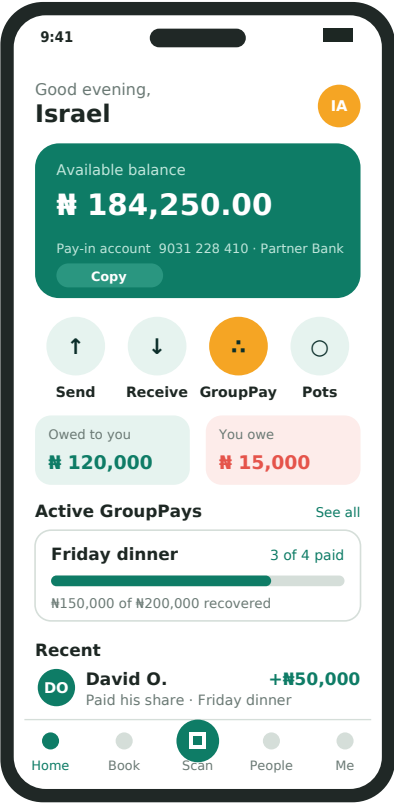
E8. Key API endpoints (REST, JSON)

Method + path	Purpose
POST /v1/auth/otp · POST /v1/auth/verify	Start and complete sign-in
POST /v1/kyc/tier1 · POST /v1/kyc/upgrade	Submit KYC data
GET /v1/wallet · GET /v1/transactions?cursor=	Balance and history
POST /v1/name-enquiry	Resolve bank account to verified name
POST /v1/payments (Idempotency-Key header required)	Send money
POST /v1/grouppays · GET /v1/grouppays/{id}	Create and view GroupPay
POST /v1/grouppays/{id}/nudges · POST /v1/grouppays/{id}/close	Remind and close
GET /p/{share_token} (public)	Guest checkout page data
POST /v1/ious · POST /v1/ious/{id}/confirm · /dispute	IOU lifecycle
POST /v1/events · POST /v1/events/{id}/spray	Owambe Spray
POST /hooks/partner/credit · /hooks/partner/transfer · /hooks/whatsapp	Inbound webhooks (signature-verified, idempotent)

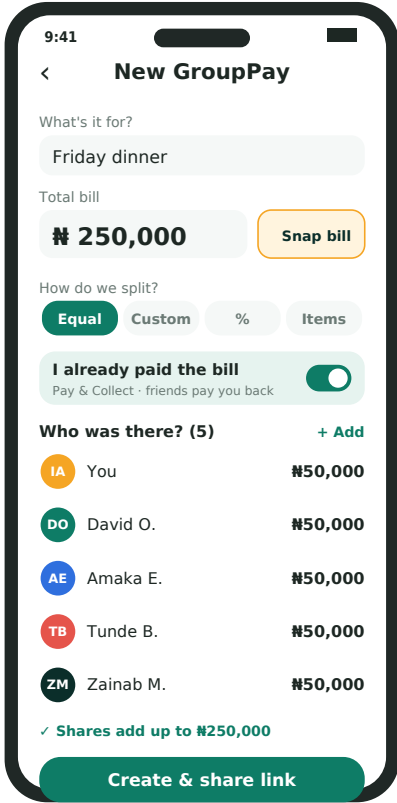
PART F

UI designs

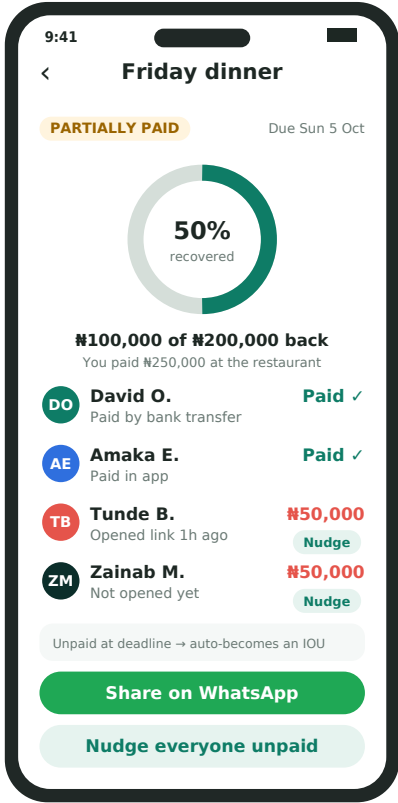
Nine core screens. The design goal is that a first-time user understands what to do in under a minute, and an invitee can pay without ever seeing MoniWise jargon. Money owed to you is green, money you owe is coral, warnings are coral boxes.



1. Home

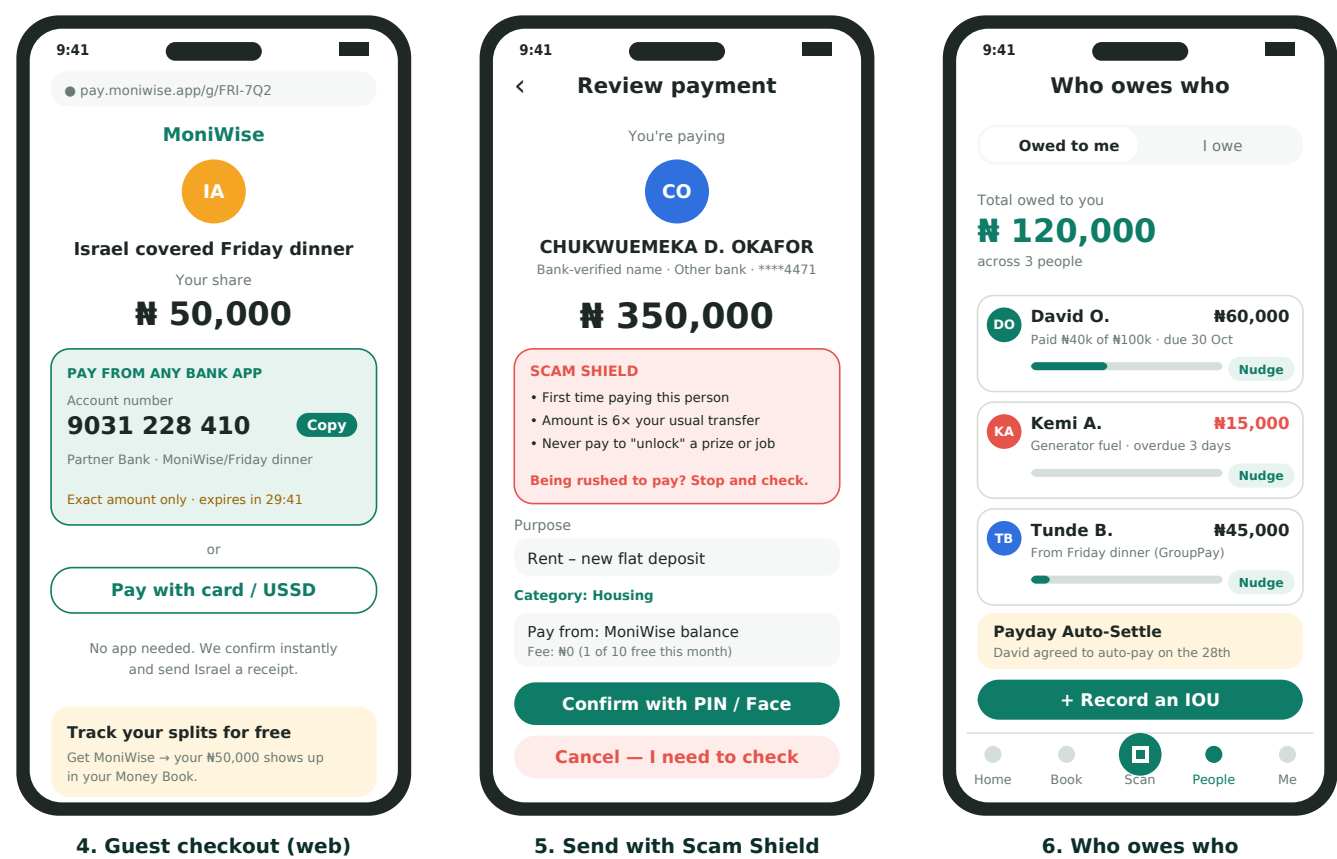


2. New GroupPay

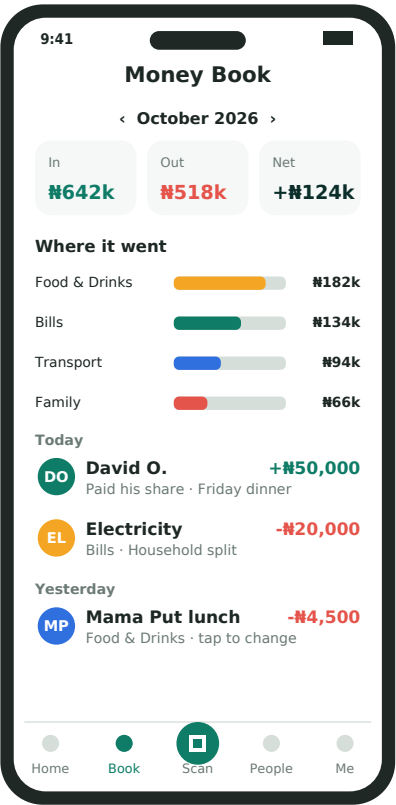


3. GroupPay live status

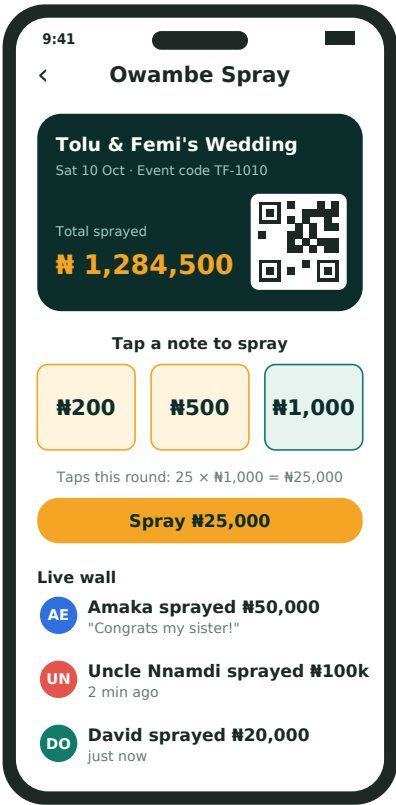
Screen	Design decisions
1. Home	GroupPay gets the accent colour; "Owed to you / You owe" are always visible because they drive return visits; pay-in account number is one tap to copy.
2. New GroupPay	One screen, no wizard. Bill Snap sits beside the amount. Pay & Collect is a single toggle with plain-language explanation. Live "adds up" check prevents errors.
3. Live status	Progress ring answers "how much is back?" at a glance. Each unpaid row has its own Nudge. The note about IOUs reassures the organiser that nothing is lost.



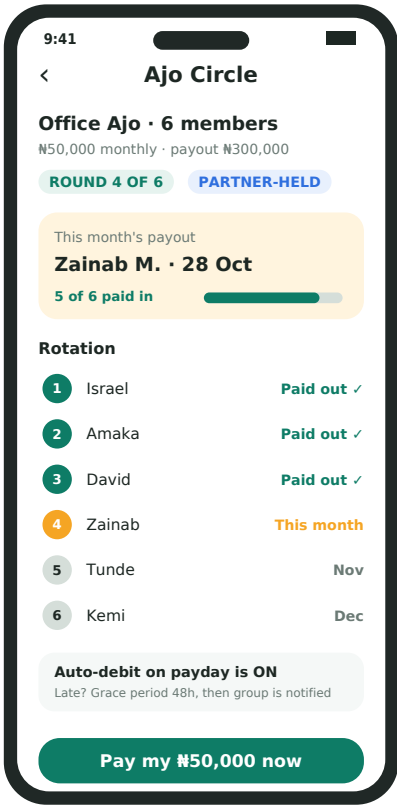
Screen	Design decisions
4. Guest checkout	Runs in any browser from WhatsApp. Account number is the hero; card/USSD is secondary. The install prompt comes after value, never before.
5. Scam Shield	Bank-verified name in capitals, reasons in plain words, and "Cancel" styled as a real choice. Purpose and category captured here feed Money Book.
6. Who owes who	Owed-to-me and I-owe tabs; progress bars on partial payments; overdue items in coral; Payday Auto-Settle shown as an agreement, not a penalty.



7. Money Book



8. Owambe Spray



9. Ajo Circle (launch window)

Screen	Design decisions
7. Money Book	Every row keeps its human reason ("David paid his share · Friday dinner"). Tap to recategorise; In/Out/Net summary on top.
8. Owambe Spray	Dark celebratory card, big QR for the venue screen, tap-to-spray notes with a multiplier, and a live wall that makes generosity visible.
9. Ajo Circle	Rotation order and who is next are always visible; "Partner-held" badge builds trust; grace period rules are stated up front.

F2. Design system


Primary green

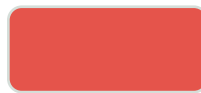
#0E7C66 · Actions, money in


Deep

#0B2E2A · Headings, dark cards


Amber

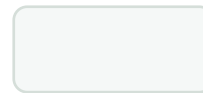
#F5A524 · GroupPay, celebration


Coral

#E5444B · You owe, warnings


Info blue

#2F6FDE · Partners, info


Surface

#F5F8F7 · Backgrounds

Token	Value	Rule
Typeface	Inter (or similar humanist sans), tabular figures for amounts	Amounts always use tabular numerals so columns align
Type scale	28 / 20 / 16 / 14 / 12	Balance 28 bold; body 14; captions 12. Never below 12 in production
Spacing	4-point grid; screen padding 16	Touch targets at least 44 x 44
Corners	Cards 16, buttons 24 (pill), chips 12	Rounded and friendly, not playful
Money colours	Green = to you, Coral = from you	Colour is never the only signal; always add + / - or words
Accessibility	WCAG 2.2 AA contrast; screen-reader labels on every amount	Read amounts as words ("fifty thousand naira")
Motion	150-250 ms; one celebration moment (Settled, Spray)	Respect reduce-motion setting
Languages	English at launch; Pidgin next; Yoruba, Igbo, Hausa later	All strings in a translation file from day one

F3. UX copy rules

Instead of...	Say...
Beneficiary	Who are you paying?
Debit account	Pay from
Transaction successful	David has been paid
Outstanding receivable	David still owes you ₦30,000
Transaction pending	We're confirming with the bank. This usually takes under a minute.
Payment reminder	"Hi Tunde, quick one: your ₦50,000 for Friday dinner is still open. Pay here: link" (sent by MoniWise, not Israel)
Error 503	The bank is slow right now. Your money is safe. Try again in a few minutes.

PART G

Build plan

G1. Launch team (9-10 people)

Role	Count	Owns
Founder / product lead	1	Vision, partners, fundraising, final scope calls
Tech lead / CTO	1	Architecture, ledger, security, partner integrations
Backend engineers	2-3	Payments, GroupPay, IOU, ledger, reconciliation, admin
Mobile engineers (Flutter)	2	App; one also builds guest web checkout
Product designer	1	Research, UI, copy, design system
QA engineer	1	Test plans from FRD acceptance criteria; automation
Compliance, risk and ops lead	1	Partner compliance, rules, reconciliation exceptions, support at launch
External	as needed	Payments counsel, penetration testers, accounting adviser

G2. Sprint plan (2-week sprints)

Sprint	Backend	Mobile / web	Done when
1-2	Repo, CI/CD, environments, ledger core, chart of accounts	Design system, navigation, onboarding UI	Ledger passes balance tests
3-4	Auth, devices, KYC provider, partner wallet sandbox	Sign-up, KYC, Home	User reaches Tier 1 in sandbox
5-6	Payments orchestrator, name enquiry, risk rules v1, webhooks	Send, Receive, Scam Shield	Money in and out end to end
7-8	GroupPay, shares, virtual accounts, matching, deadlines → IOU	New GroupPay, live status, Bill Snap	Flow E2 works end to end
9-10	WhatsApp BSP, nudges, IOU, Money Book, Spray	Guest checkout, IOU, Money Book, Spray	All P0 features complete
11	Reconciliation jobs, admin console, maker-checker	Polish, accessibility, translations file	Pen test started
12	Hardening, load test to 200 TPS, DR drill	Beta build to stores (testing tracks)	Closed beta live
13-14	Beta fixes, monitoring tuning	Beta fixes, store listings	Go-live checklist green

G3. Go-live checklist

- Signed partner agreement(s) and written counsel opinion covering wallet, virtual accounts, collections and Spray.
- Name, trademark and domain cleared; app store listings approved.
- Pen test: no open critical or high findings. Mobile app passes MASVS L1.
- Ledger reconciles to the partner pool account to the kobo for 14 consecutive beta days.
- Load test at 2× expected peak; DR failover drill completed within RTO.
- Risk rules, limits and new-organisier caps configured and signed off by the compliance lead.
- Support playbooks for the top 15 issues; in-app support live; status page ready.
- Data protection: privacy notice, consent screens, retention schedule, breach procedure.

G4. Open decisions for the founder

Decision	Options	Recommendation
Primary BaaS partner	Shortlist 3 licensed banks/providers offering dynamic virtual accounts, name enquiry and webhooks	Pick on virtual-account reliability and webhook latency, not price
Final brand name	MoniWise vs two backups	Decide after clearance in month 0
Free transfer quota	5, 10 or 20 per month	Start at 10; test elasticity in beta
Launch city	Lagos only vs Lagos + Abuja	Lagos only; concentrate social density
Plus subscription timing	At launch vs +60 days	+60 days, once usage patterns are clear

Closing note

The MVP succeeds if people say "**put it on MoniWise**" the next time a bill arrives. Every item in this document either makes that moment faster, makes it safer, or makes sure the money owed is never forgotten.